than it is now, it certainly was still an extremely cold part of the earth. It was also dark for long periods of the year.

"What's really fascinating," says Tom Rich, "is that reptiles are just not happy in cold climates. The very few that live under these conditions burrow into the mud. So how did the dinosaurs do it? What did they live on? If they were so unable to adapt, then what were they doing in that sort of climate?"

Although dinosaurs could grow to the size of a two-storey building, they also came in smaller sizes, right down to about 15 centimetres — the size of a bantam chicken. At the Dinosaur Cove digging site, next to the Otway National Park near Apollo Bay, the Richs and several volunteers unearthed the bones of a small dinosaur that was discovered to have a broken leg.

"This was a fascinating discovery, because the dinosaur lived for some time after the leg was broken. It belongs to a family of dinosaurs which was bipedal and not unlike the kangaroo. But it must have moved, or behaved in a different way to other bipedal animals, otherwise it would have ended up starving to death, and being eaten by a predator."

The number and kind of dinosaur remains found here has contributed to the radical change in scientific theories surrounding these highly misunderstood creatures. The record here now indicates not only that they had adapted successfully to cold, dark climates, thus strengthening the case that they may have been warm-blooded; it also shows that, just like mammals, they lived successfully all over the earth's surface.

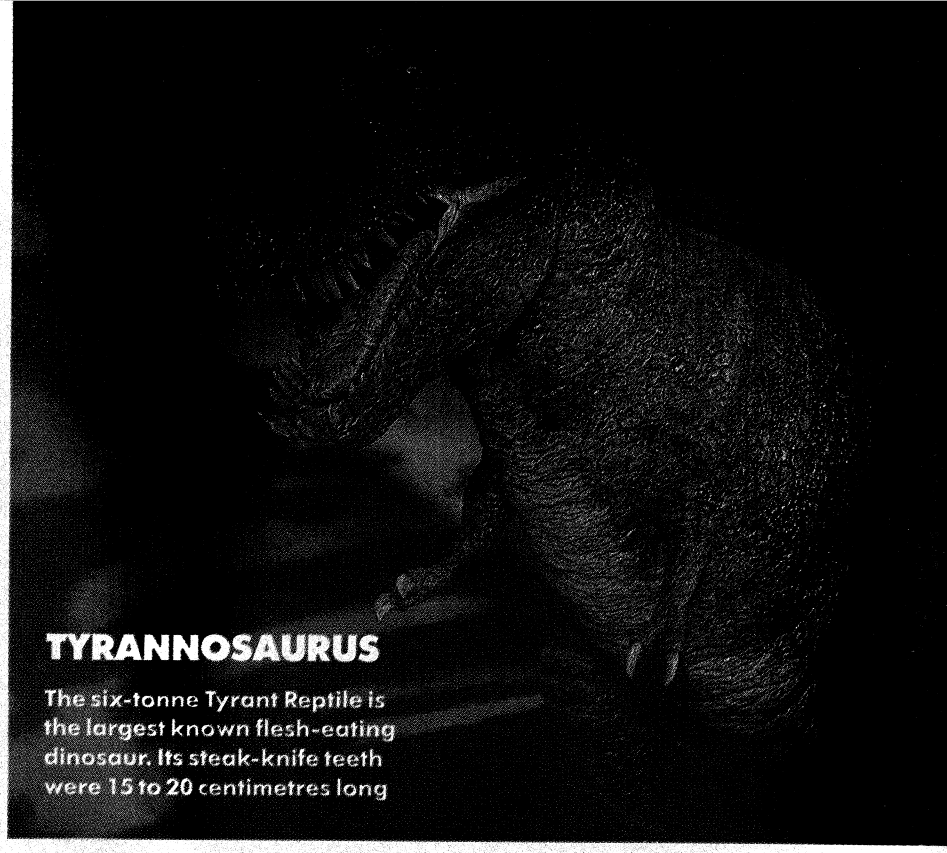
Another reason for the international attention being directed towards Victoria is the fact that one of the largest dinosaurs, *Mamenchisaurus*, is being displayed for the first time, mounted. The cast of the skeleton is 22 metres high and was originally found in China. The Victorian museum with its high ceiling is one of few places where such an experiment can be tried.

The *Mamenchisaurus* has often been depicted in drawings, rearing up on its hind legs as though browsing the top branches of a tree. No museum, however, has previously mounted a skeleton in this pose. Whether the *Mamenchisaurus* actually could browse in this manner is a point of dispute among scientists, but Tom Rich thinks it is a distinct possibility.

"When mounting it, we found some thickening around the pelvic area, which may indicate that this part of the body was meant to take some weight. One of the reasons we are exhibiting this specimen in this manner is to test the plausibility of such a mode of behaviour. It would seem to make sense that the higher a creature can reach, the more chance it has of adapting to various conditions, though we don't know if this is the case."

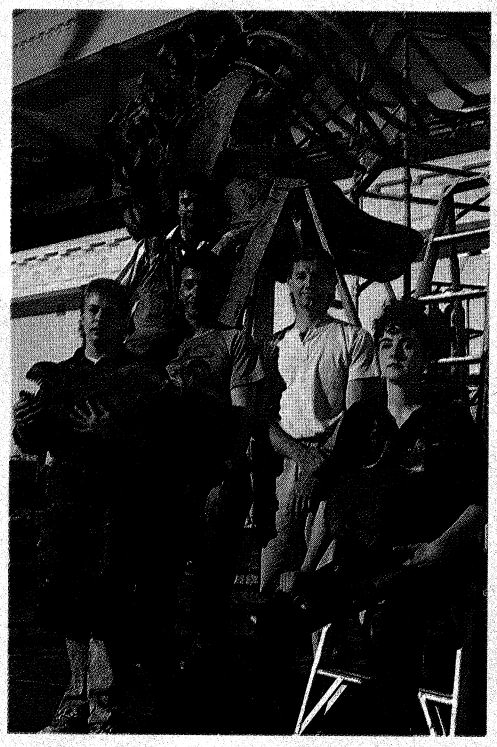
The museum's exhibition will come alive with robotic dinosaurs, half-size models of the "terrible lizards". It will include the familiar *Brontosaurus, Tyrannosaurus,* and *Stegosaurus*, and also feature lesser-known species.

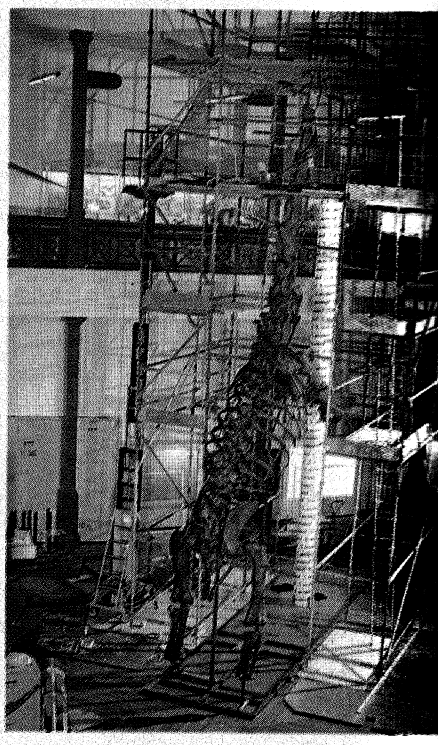
Each of these models will be powered by very pure compressed air acting through a large number of valves and pistons under the control of individual computers. Sounds are electronically reproduced, and synchronised with mouth movements.



TYRANNOSAURUS

The six-tonne Tyrant Reptile is the largest known flesh-eating dinosaur. Its steak-knife teeth were 15 to 20 centimetres long





JAMES BODDINGTON

Left, the dinosaur-building team is Dea Smith, Greg Bradshaw, Peter Swinkels, Cozens and Lissa Johnson. Right, Mamenchisaurus will stand erect for the first time

Why the world is continually fascinated by dinosaurs, while ignoring other extinct species, is a question often asked. "I really can't tell you the answer to that one," says Tom Rich, "but I can tell you that there are very few children who can't hold a reasonably intelligent discussion with you on dinosaurs."

More than 250,000 Victorians passed through each of the museum's two previous exhibitions. Even more can be expected this time, as the discoveries keep changing what we once held to be the truth. Meanwhile, consider the birds that can be seen every day. Strange as it may seem, they are the closest living relatives of the dinosaurs. ●

Dinosaurs Alive! opens at the Museum of Victoria next Sunday and continues until February.